There are six problems in total. You must solve the first five, and problem 6 is optional.

Problem 1 (CO5): Pumping Lemma (5 points)

Consider the following language.

$$L = \{0^n 1^m : n = 4q + r_1, m = 4q + r_2, q \geq 0 \text{ and } 0 \leq r_1, r_2 \leq 3\}$$

Prove that L is not regular using the pumping lemma.

Problem 2 (CO3): Context-Free Grammars (15 points)

Consider the following languages.

$$L_1 = \{w \in \{\mathtt{a}, \mathtt{b}\}^* : w \text{ is a palindrome}\}$$
$$L_2 = \{w \in \{\mathtt{a}, \mathtt{b}\}^* : \text{every second letter in } w \text{ is } \mathtt{a}\}$$
$$L_3 = \{w \in \{\mathtt{a}, \mathtt{b}\}^* : w \text{ contains more b than a}\}$$
$$L_4 = L_1 \cap L_2$$
$$L_5 = L_2 \cap L_3$$

Now solve the following problems.

(a) **Give** a context-free grammar that generates $\overline{L}_1$. (3 points) [Recall: $\overline{L}_1$ denotes the complement of the language L_1 i.e., $\overline{L}_1 = \Sigma^* - L_1$]

(b) **Give** a context-free grammar that generates L_2. (3 points)

(c) **Give** a context-free grammar that generates L_3. (3 points)

(d) **Write** all five-letter strings in L_4. (1 point)

(e) **Give** a context-free grammar that generates L_4. (2 points)

(f) **Give** a context-free grammar that generates L_5. (2 points)

(g) **Find** the total number of strings that belong to $L_5 \setminus L_1$. (1 point) [Recall: $L_5 \setminus L_1$ contains all strings that are in L_5 but not in L_1.]

Problem 3 (CO3): Derivations, Parse Trees and Ambiguity (10 points)

Consider the following context-free grammar.

$$S \to P0Q$$
$$P \to 0P0 \mid 0P1 \mid 1P0 \mid 1P1 \mid 0 \mid 1$$
$$Q \to 0Q1 \mid 1Q0 \mid \varepsilon$$

(a) **Give** a leftmost derivation for the string '0110101010'. (3 points)

(b) **Draw** the parse tree corresponding to the derivation you gave in (a). (2 points)

(c) **Show** that the given grammar is ambiguous by drawing two more parse trees (apart from the one that you have drawn in (b)) for the given string in (a). (4 points)

(d) **Write** two strings of length four that have exactly one parse tree in the given grammar. (1 point)

Problem 4 (CO3): Pushdown Automata (10 points)

Consider the following languages.

$$L_1 = \{w \in \{0,1\}^* : w \text{ starts with } 1 \text{ and ends with } 0\}$$

$$L_2 = \{w = 0^i 1^j 2^k : i, j, k \geq 0 \text{ and } i = j + 2k\}$$

$$L_3 = \{w_1 \# w_2 : w_1, w_2 \in \{0,1\}^* \text{ and the number of 1s in } w_1 \text{ is equal to the number of 0s in } w_2\}$$

Now solve the following problems.

(a) **Give** the state diagram of a PDA that recognizes L_1. (3 points)

(b) **Give** the state diagram of a PDA that recognizes L_2. (3 points)

(c) **Give** the state diagram of a PDA that recognizes L_3. (4 points)

Problem 5 (CO4): Turing Machines (10 points)

Consider the following languages.

$$L_1 = \{w_1 \# w_2 : w_1, w_2 \in \{0,1\}^* \text{ and } \text{count}(w_1, 1) = \text{count}(w_2, 1)\}$$

$$L_2 = \left\{w \# w^R : w \in \{0,1\}^*\right\}$$

(a) For the configuration $10q_3 1011$ of a Turing Machine, **what** is the current position of the tape head? (1 point)

(b) **Give** the state diagram of a turing machine that decides L_1. [Note: $\text{count}(w, a)$ denotes the number of occurrences of the symbol a in the string w.] (3 points)

(c) **Give** the state diagram of a turing machine that decides L_2. [Note: w^R denotes the reverse of w, for example, if $w = 01101$, then $w^R = 10110$.] (4 points)

(d) **Prove** that the collection of Turing decidable languages is closed under intersection. (2 points)

Problem 6 (Bonus): Turing Machines (5 points)

Let $HALT_{TM} = \{\langle M, w\rangle : M \text{ is a TM and } M \text{ halts on input } w\}$. Given that $HALT_{TM}$ is undecidable, **prove** that the collection of Turing recognizable languages is not closed under complement.

After you are finished with the exam, please indicate where you stand on the smiley face spectrum.